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DISPLAY DEVICE, CONTROLLER, AND METHOD OF DRIVING DISPLAY DEVICE

Abstract

A display device includes a display panel including a plurality of pixels, a power management circuit which provides a power voltage to the plurality of pixels through a power line, a current sensing circuit which senses a panel current flowing through the power line and performs an analog-to-digital conversion for converting the panel current into a panel current code, and a controller which sets at least one of a conversion range, a conversion time, or a conversion sampling count of the analog-to-digital conversion based on a dimming value for controlling a luminance of an image displayed by the display panel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims priority under 35 USC § 119 to Korean Patent Application No. 10-2024-0028535 filed on Feb. 28, 2024, in the Korean Intellectual Property Office (KIPO), the entire disclosure of which is incorporated by reference herein.

BACKGROUND

1. Technical Field

[0002] Embodiments relate to a display device. More particularly, embodiments relate to a display device in which a panel current may be sensed, a controller included in the display device, and a method of driving the display device.

2. Discussion of Related Art

[0003] A display device may include a display panel and a current sensing circuit. The display panel may display an image. The current sensing circuit may sense a panel current flowing in the display panel.

[0004] When a luminance of the display panel is not adjusted based on a load of input image data, overcurrent may flow into the display panel, excess power consumption may occur, and the display panel may be damaged depending on a magnitude of the overcurrent.

SUMMARY

[0005] Embodiments provide a display device in which an analog-to-digital conversion of a panel current may be performed in consideration of a magnitude of the panel current.

[0006] Embodiments provide a controller which may set an analog-to-digital conversion of a panel current in consideration of a magnitude of the panel current.

[0007] Embodiments provide a method of driving a display device which may perform an analog-to-digital conversion of a panel current in consideration of a magnitude of the panel current.

[0008] A display device according to embodiments may include a display panel including a plurality of pixels, a power management circuit which provides a power voltage to the plurality of pixels through a power line, a current sensing circuit which senses a panel current flowing through the power line and performs an analog-to-digital conversion for converting the panel current into a panel current code, and a controller which sets at least one of a conversion range, a conversion time, or a conversion sampling count of the analog-to-digital conversion based on a dimming value for controlling a luminance of an image displayed by the display panel.

[0009] In an embodiment, the controller may set the conversion range such that the conversion range increases as the dimming value increases and the conversion range decreases as the dimming value decreases, may set the conversion time such that the conversion time decreases as the dimming value increases and the conversion time increases as the dimming value decreases, and may set the conversion sampling count such that the conversion sampling count decreases as the dimming value increases and the conversion sampling count increases as the dimming value decreases.

[0010] In an embodiment, the dimming value may include a dimming gain value for decreasing the luminance of the image in a total load section and a dimming peak value for decreasing the luminance of the image in a local load section.

[0011] In an embodiment, the controller may set the conversion range such that the conversion range increases as the dimming gain value increases and the conversion range decreases as the dimming gain value decreases.

[0012] In an embodiment, the controller may set the conversion time such that the conversion time decreases as the dimming gain value increases and the conversion time increases as the dimming gain value decreases.

[0013] In an embodiment, the controller may set the conversion sampling count such that the conversion sampling count decreases as the dimming gain value increases and the conversion sampling count increases as the dimming gain value decreases.

[0014] In an embodiment, the controller may set the conversion time such that the conversion time decreases as the dimming peak value increases and the conversion time increases as the dimming peak value decreases.

[0015] In an embodiment, the controller may set the conversion sampling count such that the conversion sampling count decreases as the dimming peak value increases and the conversion sampling count increases as the dimming peak value decreases.

[0016] In an embodiment, the controller may include a net power control block which determines a load of input image data and a scale factor based on the input image data, a net power control lookup table, and the dimming value, and a conversion setting block of the controller, which sets the at least one of the conversion range, the conversion time, or the conversion sampling count based on the dimming value, and provides the at least one of the conversion range, the conversion time, or the conversion sampling count to the current sensing circuit.

[0017] In an embodiment, the conversion setting block may include a conversion range setting block which sets the conversion range based on the dimming gain value, a conversion time setting block which sets the conversion time based on the dimming gain value and the dimming peak value, and a conversion sampling count setting block which sets the conversion sampling count based on the dimming gain value and the dimming peak value.

[0018] In an embodiment, the controller may further include a global current management block which receives the panel current code from the current sensing circuit, determines a target current code corresponding to the load of the input image data, and controls the power management circuit to supply the panel current according to a comparison of the panel current code with the target current code.

[0019] In an embodiment, the controller may further include an over-current protection block which receives the panel current code from the current sensing circuit, compares the panel current code with a reference over-current code, and controls the power management circuit to stop supplying the power voltage when the panel current code is greater than the reference over-current code.

[0020] In an embodiment, the current sensing circuit may include a conversion block which converts the panel current into the panel current code, and a setting register which stores the conversion range, the conversion time, and the conversion sampling count.

[0021] In an embodiment, the controller may be implemented as a controller integrated circuit, and the current sensing circuit may be implemented as a current sensing integrated circuit different from the controller integrated circuit.

[0022] In an embodiment, the controller may set the at least one of the conversion range, the conversion time, or the conversion sampling count by changing a value of the setting register via a bus connecting between the controller integrated circuit and the current sensing integrated circuit.

[0023] A controller of a display device according to embodiments may include a net power control block which determines a scale factor based on input image data, a net power control lookup table, and a dimming value for controlling a luminance of an image displayed by the display device, and a conversion setting block which sets at least one of a conversion range, a conversion time, or a conversion sampling count of an analog-to-digital conversion for converting a panel current into a panel current code based on the dimming value, and provides the at least one of the conversion range, the conversion time, or the conversion sampling count to a current sensing circuit which senses the panel current.

[0024] In an embodiment, the conversion setting block may set the conversion range such that the conversion range increases as the dimming value increases and the conversion range decreases as the dimming value decreases, may set the conversion time such that the conversion time decreases

as the dimming value increases and the conversion time increases as the dimming value decreases, and may set the conversion sampling count such that the conversion sampling count decreases as the dimming value increases and the conversion sampling count increases as the dimming value decreases.

[0025] In an embodiment, the controller may be implemented as a controller integrated circuit, and the current sensing circuit may be implemented as a current sensing integrated circuit different from the controller integrated circuit. The conversion setting block may provide the at least one of the conversion range, the conversion time, or the conversion sampling count to the current sensing circuit via a bus connecting between the controller integrated circuit and the current sensing integrated circuit.

[0026] A method of driving a display device according to embodiments may include setting at least one of a conversion range, a conversion time, and a conversion sampling count of an analog-to-digital conversion for converting a panel current into a panel current code based on a dimming value for controlling a luminance of an image displayed by the display device, sensing a panel current, and performing the analog-to-digital conversion for converting the panel current into the panel current code.

[0027] In an embodiment, the setting may include at least one of setting the conversion range such that the conversion range increases as the dimming value increases and the conversion range decreases as the dimming value decreases, setting the conversion time such that the conversion time decreases as the dimming value increases and the conversion time increases as the dimming value decreases, or setting the conversion sampling count such that the conversion sampling count decreases as the dimming value increases and the conversion sampling count increases as the dimming value decreases.

[0028] In a display device, a controller, and a method of driving the display device according to embodiments, at least one of the conversion range, the conversion time, or the conversion sampling count of the analog-to-digital conversion for converting the panel current into the panel current code may be set based on the dimming value, and a current oscillation of the panel current may decrease when the panel current is low, and power consumption of the display device due to a rush current may decrease when the panel current is high.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] Illustrative, non-limiting embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings.

[0030] FIG. 1 is a block diagram showing a display device according to an embodiment.

[0031] FIG. 2 is a circuit diagram showing an example of a pixel of FIG. 1.

[0032] FIG. 3 is a view showing a panel current for each conversion range of an analog-to-digital conversion.

[0033] FIG. 4 is a view showing a panel current for each conversion time of an analog-to-digital conversion.

[0034] FIG. 5 is a view showing a panel current for each conversion sampling count of an analog-to-digital conversion.

[0035] FIG. 6 is a view showing a panel current for each conversion time of an analog-to-digital conversion when a rush current occurs.

[0036] FIG. 7 is a graph showing a scale factor for each dimming gain value.

[0037] FIG. 8 is a graph showing a panel current for each dimming gain value.

[0038] FIG. 9 is a graph showing a scale factor for each dimming peak value.

[0039] FIG. 10 is a graph showing a panel current for each dimming peak value.

[0040] FIG. **11** is a graph showing a relationship between a dimming gain value and setting values of an analog-to-digital conversion.

[0041] FIG. **12** is a graph showing a relationship between a dimming peak value and setting values of an analog-to-digital conversion.

[0042] FIG. **13** is a table showing an example of setting values of an analog-to-digital conversion according to a dimming gain value and a dimming peak value.

[0043] FIG. **14** is a block diagram showing a current sensing circuit and a controller according to an embodiment.

[0044] FIG. **15** is a flowchart showing a method of driving a display device according to an embodiment.

[0045] FIG. **16** is a block diagram showing an electronic apparatus according to an embodiment.

DETAILED DESCRIPTION

[0046] Hereinafter, a display device, a controller, and a method of driving a display device according to embodiments of the present disclosure will be described in more detail with reference to the accompanying drawings. Inventive concepts may be implemented in various modifications and have various forms. It is to be understood, however, that the inventive concepts are not intended to be limited to the particular forms disclosed, but on the contrary, is intended to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the inventive concepts. The same or similar reference numerals will be used for the same elements in the accompanying drawings. In the drawings, the thicknesses, the ratios, and the dimensions of the elements may be exaggerated for the effective description of the technical contents.

[0047] According to an embodiment, a current sensing circuit may perform analog-to-digital conversion for converting an analog panel current into a digital panel current. The display device may perform operations, including a global current management (GCM) operation or an over-current protection operation based on the analog panel current converted into the digital panel current. Further, the display device may detect a rush current flowing in the display panel based on the digital panel current.

[0048] FIG. **1** is a block diagram showing a display device **100** according to an embodiment. FIG. **2** is a circuit diagram showing an example of a pixel PX of FIG. **1**. FIG. **3** is a view showing a panel current for each conversion range CR of an analog-to-digital conversion. FIG. **4** is a view showing a panel current for each conversion time CT of an analog-to-digital conversion. FIG. **5** is a view showing a panel current for each conversion sampling count CSC of an analog-to-digital conversion. FIG. **6** is a view showing a panel current for each conversion time CT of an analog-to-digital conversion when a rush current occurs. FIG. **7** is a graph showing a scale factor SF for each dimming gain value DIM_GN. FIG. **8** is a graph showing a panel current PC for each dimming gain value DIM_GN. FIG. **9** is a graph showing a scale factor SF for each dimming peak value DIM_PK. FIG. **10** is a graph showing a panel current PC for each dimming peak value DIM_PK. FIG. **11** is a graph showing a relationship between a dimming gain value DIM_GN and setting values CR, CT, and CSC of an analog-to-digital conversion. FIG. **12** is a graph showing a relationship between a dimming peak value DIM_PK and setting values CR, CT, and CSC of an analog-to-digital conversion. FIG. **13** is a table showing an example of setting values CR, CT, and CSC of an analog-to-digital conversion according to a dimming gain value DIM_GN and a dimming peak value DIM_PK. FIG. **14** is a block diagram showing a current sensing circuit **150** and a controller **170** according to an embodiment. FIG. **15** is a flowchart showing a method of driving a display device according to an embodiment. FIG. **16** is a block diagram showing an electronic apparatus **1100** according to an embodiment.

[0049] Referring to FIG. **1**, a display device **100** according to an embodiment may include a display panel **110**, a data driver **120**, a scan driver **130**, a power management circuit **140**, a current sensing circuit **150**, and a controller **170**.

[0050] The display panel **110** may include a plurality of pixels PX. The display panel **110** may

display an image. For example, the display panel **110** may display an image through light emitted from each of the pixels PX.

[0051] Referring to FIG. 2, the pixel PX may include a first transistor T1, a second transistor T2, a storage capacitor CST, and a light-emitting element EL.

[0052] The first transistor T1 may include a gate connected to a second electrode of the second transistor T2 and a first electrode of the storage capacitor CST, a first electrode connected to a power line PSL transmitting a first power voltage ELVDD, and a second electrode connected to a second electrode of the storage capacitor CST and an anode of the light-emitting element EL.

[0053] The second transistor T2 may include a gate receiving a scan signal SC, a first electrode connected to a data line DL transmitting a data signal DS, and the second electrode connected to the gate of the first transistor T1 and the first electrode of the storage capacitor CST.

[0054] The storage capacitor CST may include the first electrode connected to the gate of the first transistor T1 and the second electrode of the second transistor T2, and the second electrode connected to the second electrode of the first transistor T1 and the anode of the light-emitting element EL.

[0055] The light-emitting element EL may include the anode connected to the second electrode of the first transistor T1 and the second electrode of the storage capacitor CST, and a cathode receiving a second power supply voltage ELVSS.

[0056] In an embodiment, the light-emitting element EL may be an organic light-emitting diode (OLED). In an embodiment, the light-emitting element EL may be an inorganic light-emitting diode, a micro light-emitting diode, or a quantum-dot light-emitting diode (QLED).

[0057] Referring to FIG. 1, the data driver **120** may supply data signals DS to the pixels PX of the display panel **110**. The data driver **120** may generate the data signals DS based on output image data ODAT and a data control signal DCTRL. The output image data ODAT may include grayscale values corresponding to the pixels PX. The data control signal DCTRL may include an output data enable signal, a horizontal start signal, or a load signal. Embodiments are not limited thereto, and the data control signal DCTRL may include another signal.

[0058] The scan driver **130** may supply scan signals SC to the pixels PX of the display panel **110**. The scan driver **130** may generate the scan signals SC based on a scan control signal SCTRL. The scan control signal SCTRL may include a scan start signal or a scan clock signal. Embodiments are not limited thereto, and the scan control signal SCTRL may include another signal.

[0059] The power management circuit **140** may supply the first power voltage ELVDD to the pixels PX of the display panel **110** through the power line PSL. The power management circuit **140** may generate the first power voltage ELVDD based on a power control signal VC. The power management circuit **140** may be implemented as a power management integrated circuit.

[0060] The current sensing circuit **150** may sense a panel current PC flowing through the power line PSL. The panel current PC may be a global current. The panel current PC may be a global current that is the sum of driving currents flowing through the light-emitting elements EL of the pixels PX. The current sensing circuit **150** may perform an analog-to-digital conversion. The current sensing circuit **150** may convert the panel current PC into a panel current code PCC. Specifically, the analog-to-digital conversion may include a sampling operation that may sample a sensing voltage corresponding to the panel current PC and a quantization and encoding operation that may convert the sampled sensing voltage into the panel current code PCC.

[0061] The current sensing circuit **150** may receive at least one of a conversion range CR, a conversion time CT, or a conversion sampling count CSC. For example, the current sensing circuit **150** may be set using the conversion range CR, the conversion time CT, and the conversion sampling count CSC of the analog-to-digital conversion, and the current sensing circuit **150** may perform the analog-to-digital conversion based on the conversion range CR, the conversion time CT, and the conversion sampling count CSC.

[0062] The conversion range CR may indicate a range of the sensing voltage to be sampled. As the

conversion range CR increases, a wider range of the panel current PC may be converted into the panel current code PCC. As the conversion range CR increases, a resolution of the panel current PC converted into the panel current code PCC may decrease, and quantization noise may increase during an analog-to-digital conversion. In other words, as the conversion range CR decreases, the resolution of the panel current PC converted into the panel current code PCC may increase, and the quantization noise may decrease during an analog-to-digital conversion. A first line **210** of FIG. **3** may be the panel current indicated by the panel current code PCC when the conversion range CR is about ± 163.84 mV, and a second line **220** of FIG. **3** may be the panel current indicated by the panel current code PCC when the conversion range CR is about ± 40.96 mV. The quantization noise of an analog-to-digital conversion when the conversion range CR is about ± 40.96 mV may be less than the quantization noise of an analog-to-digital conversion when the conversion range CR is about ± 163.84 mV, and a current oscillation of the panel current shown by the second line **220** indicated by the panel current code PCC when the conversion range CR is about ± 40.96 mV may be less than a current oscillation of the panel current shown by the first line **210** indicated by the panel current code PCC when the conversion range CR is about ± 163.84 mV.

[0063] The conversion time CT may indicate a time taken to convert the sampled sensing voltage into the panel current code PCC. In other words, the conversion time CT may indicate a time for the quantization and encoding operation. As the conversion time CT increases, the quantization noise may decrease in the analog-to-digital conversion. A third line **310** of FIG. **4** may be the panel current indicated by the panel current code PCC when the conversion time CT is about 50 microseconds (us), and a fourth line **320** of FIG. **4** may be the panel current indicated by the panel current code PCC when the conversion time CT is about 140 us. The quantization noise of the analog-to-digital conversion when the conversion time CT is about 140 us may be less than the quantization noise of the analog-to-digital conversion when the conversion time CT is about 50 us, and a current oscillation of the panel current shown by the fourth line **320** indicated by the panel current code PCC when the conversion time CT is about 140 us may be less than a current oscillation of the panel current shown by the third line **310** indicated by the panel current code PCC when the conversion time CT is about 50 us.

[0064] The conversion sampling count CSC may indicate a number of times sampling operations of the sensing voltages are performed before the quantization and encoding operation. When the conversion sampling count CSC is 1, a sensing voltage sampled once may be converted into the panel current code PCC. When the conversion sampling count CSC is greater than 1, an average of sensing voltages sampled multiple times may be converted into the panel current code PCC. In an embodiment, as the conversion sampling count CSC increases, the quantization noise in the analog-to-digital conversion may decrease. A fifth line **410** of FIG. **5** may be the panel current indicated by the panel current code PCC when the conversion sampling count CSC is 1, and a sixth line **420** of FIG. **5** may be the panel current indicated by the panel current code PCC when the conversion sampling count CSC is 16. The quantization noise of the analog-to-digital conversion when the conversion sampling count CSC is 16 may be less than the quantization noise of the analog-to-digital conversion when the conversion sampling count CSC is 1, and a current oscillation of the panel current shown by the sixth line **420** indicated by the panel current code PCC when the conversion sampling count CSC is 16 may be less than a current oscillation of the panel current shown by the fifth line **410** indicated by the panel current code PCC when the conversion sampling count CSC is 1.

[0065] When the panel current PC is low, as a ratio of the magnitude of the current oscillation of the panel current PC to the magnitude of the panel current PC may increase, and an influence of the current oscillation of the panel current PC may be high. When the display panel **110** displays a low-grayscale image, the panel current PC may be low, and flicker due to the current oscillation of the panel current PC may be recognized in the low-grayscale image. Further, when the panel current PC is low in a global current management (GCM) operation and an over-current protection (OCP)

operation performed based on the panel current PC, the current oscillation of the panel current PC may be reduced as described herein.

[0066] The display device **100** may detect a rush current generated in the display panel **110** through the panel current PC sensed by the current sensing circuit **150**. A rush current may be a current of the display panel **110** that is greater than a rush reference current. The rush reference current may be a current level associated with an increased level of power consumption of the display device **100**. The rush reference current may be predetermined current level. When the panel current PC becomes greater than the rush reference current, an alert signal may be generated, and the power management circuit **140** may decrease the first power voltage ELVDD in response to the alert signal, and the power consumption of the display device **100** may be reduced.

[0067] As the conversion range CR decreases, a narrow range of panel current PC may be converted into the panel current code PCC, and as the conversion range CR decreases, limitations may arise in detecting a rush current through the panel current PC.

[0068] As the conversion time CT increases, a time taken to convert the panel current PC into the panel current code PCC may increase, and a response time to a rush current may increase. Further, as the conversion time CT increases, the power consumption of the display device **100** due to a rush current may increase. A first plot **510** of FIG. **6** may be the panel current indicated by the panel current code PCC when the conversion time CT is about 80 us and a second plot **520** of FIG. **6** may be the panel current indicated by the panel current code PCC when the conversion time CT is about 320 us. A first signal **530** of FIG. **6** may be a pulse of the alert signal ALERT when the conversion time CT is about 80 us, and a second signal **540** of FIG. **6** may be a pulse of the alert signal ALERT when the conversion time CT is about 320 us. A first response time RT1 of FIG. **6** may be the response time to a rush current when the conversion time CT is about 80 us, and a second response time RT2 of FIG. **6** may be the response time to a rush current when the conversion time CT is about 320 us. The second response time RT2 to a rush current when the conversion time CT is about 320 us may be greater than the first response time RT1 to a rush current when the conversion time CT is about 80 us. A maximum value of the second plot **520** of the panel current, which may be a rush current, when the conversion time CT is about 320 us may be greater than a maximum value of the first plot **510** of the panel current, which may also be a rush current, when the conversion time CT is about 80 us.

[0069] As the conversion sampling count CSC increases, a time taken to convert the panel current PC into the panel current code PCC may increase, and the response time to a rush current may increase. Further, as the conversion sampling count CSC increases, the power consumption of the display device **100** due to a rush current may increase.

[0070] When the panel current PC is high, as the power consumption of the display device **100** due to a rush current increases, the influence of the response time to a rush current may be high. Further, when the panel current PC is high, as the ratio of the magnitude of the current oscillation of the panel current PC to the magnitude of the panel current PC decreases, the influence of the current oscillation of the panel current PC may be low.

[0071] The current sensing circuit **150** may be implemented as a current sensing integrated circuit **160**. The current sensing circuit **150** may be implemented as a current sensing integrated circuit **160** different from the power management circuit **140**. For example, the power management circuit **140** and the current sensing integrated circuit **160** may be implemented as distinct circuits. In an example, the power management circuit **140** and the current sensing integrated circuit **160** may be implemented in different chips.

[0072] The controller **170** may control an operation of the data driver **120**, an operation of the scan driver **130**, and an operation of the power management circuit **140**. The controller **170** may supply the output image data ODAT and the data control signal DCTRL to the data driver **120**. The controller **170** may generate the output image data ODAT based on input image data IDAT. The input image data IDAT may include grayscale values corresponding to the pixels PX. The

controller **170** may generate the data control signal DCTRL based on a control signal CTRL. The control signal CTRL may include a vertical synchronization signal, a horizontal synchronization signal, an input data enable signal, and a master clock signal. Embodiments are not limited thereto, and the control signal CTRL may include another signal.

[0073] The controller **170** may supply the scan control signal SCTRL to the scan driver **130**. The controller **170** may generate the scan control signal SCTRL based on the control signal CTRL.

[0074] The controller **170** may supply the power control signal VC to the power management circuit **140**. The controller **170** may generate the power control signal VC based on the control signal CTRL and the power current code PCC.

[0075] The controller **170** may be implemented as a controller integrated circuit **180** different from a power management integrated circuit and the current sensing integrated circuit **160**.

[0076] The controller **170** may provide at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC to the current sensing circuit **150**. For example, the controller **170** may provide at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC to the current sensing circuit **150** via a bus connecting between the controller integrated circuit **180** and the current sensing integrated circuit **160**. In an embodiment, the controller **170** may provide at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC to the current sensing circuit **150** via a bus connecting between the controller integrated circuit **180** and the current sensing integrated circuit **160**. For example, the bus may support an Inter-Integrated Circuit (I2C) communication. The I2C communication may use a multi-controller/multi-target, single-ended, serial communication bus.

[0077] The current sensing circuit **150** may provide the power current code PCC to the controller **170**. For example, the current sensing circuit **150** may provide the power current code PCC to the controller **170** via the I2C communication between the current sensing integrated circuit **160** and the controller integrated circuit **180**.

[0078] The controller **170** may provide the power control signal VC to the power management circuit **140**. For example, the controller **170** may provide the power control signal VC to the power management circuit **140** via the I2C communication between the controller integrated circuit **180** and the power management circuit **140**.

[0079] The controller **170** may receive a dimming value DIM from outside the display device **100**. For example, the controller **170** may receive the dimming value DIM from a host processor. The dimming value DIM may be a value set so that the display device **100** may display an image having a target luminance level.

[0080] The controller **170** may perform a net power control (NPC) operation. The controller **170** may perform a net power control (NPC) operation for controlling a luminance of an image displayed by the display panel **110** based on a load of the input image data IDAT. The controller **170** may perform the NPC operation, which may decrease the luminance of an image displayed by the display panel **110** based on the load of the input image data IDAT. The controller **170** may generate a net power control curve based on a reference net power control curve and the dimming value DIM. Each of the reference net power control curve and the net power control curve may represent a scale factor in a total load section (e.g., 0% to 100%) of an image. For example, when the input image data IDAT is a full black image, a total load value of the input image data IDAT may be 0%, and when the input image data IDAT is a full white image, the total load value of the input image data IDAT may be 100%. The scale factor may be greater than 0 and less than or equal to 1. The dimming value DIM may be a value for decreasing the luminance of the image displayed by the display panel **110**. The controller **170** may determine the load of the input image data IDAT based on the input image data IDAT, and may determine the scale factor based on the net power control curve and the load of the input image data IDAT. The load of the input image data IDAT may be greater than or equal to 0% and less than or equal to 100%. The controller **170** may generate the grayscale values of the output image data ODAT by applying the scale factor to the

grayscale values of the input image data IDAT.

[0081] The dimming value DIM may include a dimming gain value DIM_GN and a dimming peak value DIM_PK. The dimming gain value DIM_GN may be a value for controlling the luminance of the image. For example, the dimming gain value DIM_GN may be a value for controlling the luminance of the image in the total load section. As the dimming gain value DIM_GN decreases, the scale factor may decrease in the total load section. In an embodiment, as the dimming gain value DIM_GN decreases, the grayscale values of the output image data ODAT may decrease in the total load section, and the magnitude of the panel current PC may decrease in the total load section. In an embodiment, as the dimming gain value DIM_GN decreases, the influence of the current oscillation of the panel current PC in the total load section may increase. A third plot **610** of FIG. 7 is the net power control curve when the dimming gain value DIM_GN is 1, a fourth plot **620** of FIG. 7 is the net power control curve when the dimming gain value DIM_GN is 0.5, a fifth plot **710** of FIG. 8 is the panel current in the total load section when the dimming gain value DIM_GN is 1, and a sixth plot **720** of FIG. 8 is the panel current in the total load section when the dimming gain value DIM_GN is 0.5. The sixth plot **720** of the panel current in the total load section when the dimming gain value DIM_GN is 0.5 may be less than the fifth plot **710** of the panel current in the total load section when the dimming gain value DIM_GN is 1. In an embodiment, the influence of the current oscillation of the sixth plot **720** of the panel current in the total load section when the dimming gain value DIM_GN is 0.5 may be greater than the influence of the current oscillation of the fifth plot **710** of the panel current in the total load section when the dimming gain value DIM_GN is 1.

[0082] The dimming peak value DIM_PK may be a value for decreasing the luminance of the image in a local load section. The local load section may be a portion of the total load section. For example, the local load section may correspond to an object within an image being displayed by the display panel **110**. As the dimming peak value DIM_PK decreases, the scale factor may decrease in the local load section. In an embodiment, as the dimming peak value DIM_PK decreases, the grayscale values of the output image data ODAT in the local load section may decrease, and the magnitude of the panel current PC in the local load section may decrease. In an embodiment, as the dimming peak value DIM_PK decreases, the influence of the current oscillation of the panel current PC in the local load section may increase. A seventh plot **810** of FIG. 9 is the net power control curve when the dimming peak value DIM_PK is 1, and an eighth plot **820** of FIG. 9 is the net power control curve when the dimming peak value DIM_PK is 0.5. A ninth plot **910** of FIG. 10 is the panel current in the total load section when the dimming peak value DIM_PK is 1, and a tenth plot **920** of FIG. 10 is the panel current in the total load section when the dimming peak value DIM_PK is 0.5. The tenth plot **920** of the panel current in the local load section when the dimming peak value DIM_PK is 0.5 may be less than the ninth plot **910** of the panel current in the local load section when the dimming peak value DIM_PK is 1. Further, the influence of the current oscillation of the tenth plot **920** of the panel current in the local load section when the dimming peak value DIM_PK is 0.5 may be greater than the influence of the current oscillation of the ninth plot **910** of the panel current in local low load section when the dimming peak value DIM_PK is 1.

[0083] In the display device **100** according to an embodiment, the controller **170** may set at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC of the analog-to-digital conversion based on the dimming value DIM. The magnitude of the panel current PC may change depending on the dimming value DIM. The influence of the current oscillation of the panel current PC and the power consumption of the display device **100** due to a rush current may change depending on the magnitude of the panel current PC. The controller **170** may set at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC in consideration of the influence of the current oscillation of the panel current PC when the dimming value DIM is low, and the dimming value DIM may be low when the panel current PC is low. The controller **170** may set at least one of the conversion range CR, the

conversion time CT, or the conversion sampling count CSC in consideration of the power consumption of the display device **100** due to a rush current when the dimming value DIM is high, and the dimming value DIM may be high when the panel current PC is high.

[0084] The controller **170** may set the conversion range CR such that the conversion range CR increases as the dimming value DIM increases, and the conversion range CR decreases as the dimming value DIM decreases. In a case that the panel current PC increases as the dimming value DIM increases, the conversion range CR may increase in consideration of the detection of a rush current. Further, in a case that the panel current PC decreases as the dimming value DIM decreases, the conversion range CR may decrease in consideration of the influence of the current oscillation of the panel current PC.

[0085] The controller **170** may set the conversion time CT such that the conversion time CT decreases as the dimming value DIM increases and the conversion time CT increases as the dimming value DIM decreases. Further, the controller **170** may set the conversion sampling count CSC such that the conversion sampling count CSC decreases as the dimming value DIM increases, and the conversion sampling count CSC increases as the dimming value DIM decreases. In the case that the panel current PC increases as the dimming value DIM increases, the conversion time CT and the conversion sampling count CSC may decrease in consideration of the response time to a rush current. Further, in the case that the panel current PC decreases as the dimming value DIM decreases, the conversion time CT and the conversion sampling count CSC may increase in consideration of the influence of the current oscillation of the panel current PC.

[0086] The controller **170** may set the conversion range CR such that the conversion range CR increases as the dimming gain value DIM_GN increases and the conversion range CR decreases as the dimming gain value DIM_GN decreases, as shown in FIG. **11**. In the case that the panel current PC increases in the total load section as the dimming gain value DIM_GN increases, the conversion range CR may increase in consideration of the detection of a rush current. Further, in the case that the panel current PC decreases in the total load section as the dimming gain value DIM_GN decreases, the conversion range CR may decrease in consideration of the influence of the current oscillation of the panel current PC.

[0087] The controller **170** may set the conversion time CT such that the conversion time CT decreases as the dimming gain value DIM_GN increases and the conversion time CT increases as the dimming gain value DIM_GN decreases, as shown in FIG. **11**. Further, the controller **170** may set the conversion sampling count CSC such that the conversion sampling count CSC decreases as the dimming gain value DIM_GN increases and the conversion sampling count CSC increases as the dimming gain value DIM_GN decreases, as shown in FIG. **11**. In a case that the panel current PC increases in the total load section as the dimming gain value DIM_GN increases, the conversion time CT and the conversion sampling count CSC may decrease in consideration of the response time to a rush current. Further, in the case that the panel current PC decreases in the total load section as the dimming gain value DIM_GN decreases, the conversion time CT and the conversion sampling count CSC may increase in consideration of the influence of the current oscillation of the panel current PC.

[0088] The controller **170** may set the conversion time CT such that the conversion time CT decreases as the dimming peak value DIM_PK increases and the conversion time CT increases as the dimming peak value DIM_PK decreases, as shown in FIG. **12**. Further, the controller **170** may set the conversion sampling count CSC such that the conversion sampling count CSC decreases as the dimming peak value DIM_PK increases and the conversion sampling count CSC increases as the dimming peak value DIM_PK decreases, as shown in FIG. **12**. In a case that the panel current PC increases in the local load section as the dimming peak value DIM_PK increases, the conversion time CT and the conversion sampling count CSC may decrease in consideration of the response time for a rush current. Further, in the case that the panel current PC decreases in the local load section as the dimming peak value DIM_PK decreases, the conversion time CT and the

conversion sampling count CSC may increase in consideration of the influence of the current oscillation of the panel current PC.

[0089] The controller **170** may set the conversion range CR regardless of the magnitude of the dimming peak value DIM_PK, as shown in FIG. **12**. For example, the controller **170** may set the conversion range CR in consideration of the dimming gain value DIM_GN only. The dimming peak value DIM_PK may affect the scale factor and the panel current PC in the local load section, and may not affect the scale factor and the panel current PC in a high load section (e.g., a section where the load is close to 100%), unless the dimming peak value DIM_PK is set low, the dimming peak value DIM_PK may not be considered in setting the conversion range CR.

[0090] FIG. **13** shows an example of the conversion range CR, the conversion time CT, and the conversion sampling count CSC of the analog-to-digital conversion set according to the magnitude of the dimming gain value DIM_GN and the magnitude of the dimming peak value DIM_PK. The conversion range CR may be set to a high value (e.g., approximately ± 163.84 mV) regardless of the magnitude of the dimming peak value DIM_PK when the dimming gain value DIM_GN is high, and the conversion range CR may be set to a low value (e.g., approximately ± 40.96 mV) regardless of the magnitude of the dimming peak value DIM_PK when the dimming gain value DIM_GN is low. The conversion time CT may be set to a low or minimum value (e.g., 50 μ s) when the dimming gain value DIM_GN is high and the dimming peak value DIM_PK is high, the conversion time CT may be set to an intermediate value (e.g., 540 μ s) when the dimming gain value DIM_GN is high and the dimming peak value DIM_PK is low or when the dimming gain value DIM_GN is low and the dimming peak value DIM_PK is high, and the conversion time CT may be set to a high or maximum value (e.g., 4120 μ s) when the dimming gain value DIM_GN is low and the dimming peak value DIM_PK is low. The conversion sampling count CSC may be set to a low or minimum value (e.g., 1) when the dimming gain value DIM_GN is high and the dimming peak value DIM_PK is high, the conversion sampling count CSC may be set to an intermediate value (e.g., 64) when the dimming gain value DIM_GN is high and the dimming peak value DIM_PK is low or when the dimming gain value DIM_GN is low and the dimming peak value DIM_PK is high, and the conversion sampling count CSC may be set to a high or maximum value (e.g., 1024) when the dimming gain value DIM_GN is low and the dimming peak value DIM_PK is low.

[0091] In the display device **100** according to the present embodiment, the controller **170** may set at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC of the analog-to-digital conversion based on the dimming value DIM, and the current oscillation of the panel current PC may decrease when the panel current PC is low, and further the power consumption of the display device **100** due to a rush current may decrease when the panel current PC is high. Herein, the panel current PC may be considered to be high in a case that a rush current is detected.

[0092] FIG. **14** is a block diagram showing the current sensing circuit **150** and the controller **170** according to an embodiment.

[0093] Referring to FIGS. **1** and **14**, the controller **170** according to an embodiment may include a net power control block **171**, a conversion setting block **172**, a global current management block **173**, and an over-current protection block **174**.

[0094] The net power control block **171** may determine the load LD of the input image data IDAT and the scale factor SF based on the input image data IDAT, a net power control lookup table NPC_LUT, and the dimming value DIM. The reference net power control curve may be stored in the net power control lookup table NPC_LUT. The net power control block **171** may generate the net power control curve based on the reference net power control curve and the dimming value DIM, may determine the load LD of the input image data IDAT based on the input image data IDAT, and may determine the scale factor SF based on the net power control curve and the load LD of the input image data IDAT.

[0095] The conversion setting block **172** may set at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC based on the dimming value DIM. The conversion setting block **172** may provide at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC to the current sensing circuit **150**.

[0096] The conversion setting block **172** may set the conversion range CR such that the conversion range CR increases as the dimming value DIM increases and the conversion range CR decreases as the dimming value DIM decreases. The conversion setting block **172** may set the conversion time CT such that the conversion time CT decreases as the dimming value DIM increases and the conversion time CT increases as the dimming value DIM decreases. The conversion setting block **172** may set the conversion sampling count CSC such that the conversion sampling count CSC decreases as the dimming value DIM increases and the conversion sampling count CSC increases as the dimming value DIM decreases.

[0097] The conversion setting block **172** may include a conversion range setting block **172A**, a conversion time setting block **172B**, and a conversion sampling count setting block **172C**. The conversion range setting block **172A** may set the conversion range CR based on the dimming gain value DIM_GN. The conversion time setting block **172B** may set the conversion time CT based on the dimming gain value DIM_GN and the dimming peak value DIM_PK. The conversion sampling count setting block **172C** may set the conversion sampling count CSC based on the dimming gain value DIM_GN and the dimming peak value DIM_PK.

[0098] The global current management block **173** may receive the panel current code PCC from the current sensing circuit **150**. The global current management block **173** may receive the load LD from the net power control block **171**. The global current management block **173** may determine a target current code corresponding to the load LD. The global current management block **173** may compare the panel current code PCC with the target current code. The global current management block **173** may control the power control signal VC based on the comparison of the panel current code PCC with the target current code. For example, the global current management block **173** may transmit the power control signal VC to the power management circuit **140** to control the power management circuit **140** to supply the panel current PC according to a comparison of the panel current code with the target current code.

[0099] The global current management block **173** may transmit the power control signal VC to the power management circuit **140** via Inter-Integrated Circuit ("I2C") communication between the controller integrated circuit **180** and the power management circuit **140**.

[0100] The over-current protection block **174** may receive the panel current code PCC from the current sensing circuit **150**. The over-current protection block **174** may compare the panel current code PCC with a reference over-current code. The reference over-current code may be predetermined. The over-current protection block **174** may control the power management circuit **140** to stop supplying the first power voltage ELVDD when the panel current code PCC is greater than the reference over-current code.

[0101] The over-current protection block **174** may receive the power control signal VC from the global current management block **173**. The over-current protection block **174** may modify the power control signal VC, which may protect from an over-current condition. For example, the over-current protection block **174** may transmit the power control signal VC to the power management circuit **140** via Inter-Integrated Circuit ("I2C") communication between the controller integrated circuit **180** and the power management circuit **140**, and the power control signal VC, modified by the over-current protection block **174** may cause the power management circuit **140** to stop supplying the first power voltage ELVDD to the display panel **110**.

[0102] In an embodiment, the global current management block **173** may transmit the power control signal VC to the power management circuit **140** via Inter-Integrated Circuit ("I2C") communication between the controller integrated circuit **180** and the power management circuit **140**, and the over-current protection block **174** may transmit a separate signal, such as an over-

current code, which may stop the power management circuit **140** from supplying the first power voltage ELVDD to the display panel **110**.

[0103] The current sensing circuit **150** may include a conversion block **151** and a setting register **152**.

[0104] The conversion block **151** may convert the panel current PC into the panel current code PCC. The conversion block **151** may convert a sensing voltage VSENSE between opposite ends of a sensing resistor RSENSE disposed in the power line PSL into the panel current code PCC. The panel current PC that flows through the sensing resistor RSENSE and the sensing voltage VSENSE between the opposite ends of the sensing resistor RSENSE may be proportional to the panel current PC, and the conversion block **151** may convert the panel current PC into the panel current code PCC using the sensing voltage VSENSE.

[0105] The setting register **152** may store at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC.

[0106] The conversion setting block **172** of the controller **170** may set at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC by changing a value of the setting register **152** via the I2C communication between the controller integrated circuit **180** and the current sensing integrated circuit **160**.

[0107] FIG. **15** is a flowchart showing a method of driving a display device according to an embodiment.

[0108] Referring to FIGS. **1** and **15**, in a method of driving a display device **100** according to an embodiment, at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC of the analog-to-digital conversion for converting the panel current PC into the panel current code PCC may be set based on the dimming value DIM for decreasing the luminance of the image displayed by the display device **100** (S100), the panel current PC may be sensed (S200), and the analog-to-digital conversion for converting the panel current PC into the panel current code PCC may be performed (S300).

[0109] In setting at least one of the conversion range CR, the conversion time CT, or the conversion sampling count CSC based on the dimming value DIM (S100), the conversion range CR may be set such that the conversion range CR increases as the dimming value DIM increases and the conversion range CR decreases as the dimming value DIM decreases. The conversion time CT may be set such that the conversion time CT decreases as the dimming value DIM increases and the conversion time CT increases as the dimming value DIM decreases. The conversion sampling count CSC may be set such that the conversion sampling count CSC decreases as the dimming value DIM increases and the conversion sampling count CSC increases as the dimming value DIM decreases.

[0110] FIG. **16** is a block diagram showing an electronic apparatus **1100** according to an embodiment.

[0111] Referring to FIG. **16**, the electronic apparatus **1100** may include a processor **1110**, a memory device **1120**, a storage device **1130**, an input/output (“I/O”) device **1140**, a power supply **1150**, and a display device **1160**. The electronic apparatus **1100** may further include a plurality of ports capable of communicating with a video card, a sound card, a memory card, or a USB device, or communicating with other systems.

[0112] The processor **1110** may perform specific calculations or tasks. In an embodiment, the processor **1110** may be a microprocessor or a central processing unit (CPU). The processor **1110** may be connected to other components through at least one of an address bus, a control bus, or a data bus. In an embodiment, the processor **1110** may be connected to an expansion bus such as a peripheral component interconnect (PCI) bus. In an embodiment, the processor **1110** may provide the input image data (IDAT of FIG. **1**) and the control signal (CTRL of FIG. **1**) to the display device **1160**.

[0113] The memory device **1120** may store data for an operation of the electronic apparatus **1100**.

For example, the memory device **1120** may include, for example, a nonvolatile memory device such as an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), a flash memory, a phase change random access memory (PRAM), a resistance random access memory (RRAM), a nano floating gate memory (NFGM), a polymer random access memory (PoRAM), a magnetic random access memory (MRAM), or a ferroelectric random access memory (FRAM); and/or a volatile memory device such as a dynamic random access memory (DRAM), a static random access memory (SRAM), or a mobile DRAM.

[0114] The storage device **1130** may include a solid state drive (SSD), a hard disk drive (HDD), a CD-ROM, and the like. The I/O device **1140** may include, for example, an input device such as a keyboard, a keypad, a touch pad, a touch screen, or a mouse; and an output device such as a speaker or a printer. The power supply **1150** may supply a power required for the operation of the electronic apparatus **1100**. The display device **1160** may be connected to other components through the buses or other communication links. The display device **1160** may correspond to the display device **100** of FIG. 1.

[0115] A controller included in the display device **1160** may set at least one of a conversion range, a conversion time, or a conversion sampling count of an analog-to-digital conversion for converting a panel current into a panel current code based on a dimming value, and a current oscillation of the panel current may decrease when the panel current is low, and power consumption of the display device **1160** due to a rush current may decrease when the panel current is high.

[0116] A display device according to embodiments may be applied to an electronic display device. For example, the display device according to embodiments may be applied to a computer, a notebook, a mobile phone, a smart phone, a smart pad, a smart watch, a portable media player (PMP), a personal digital assistant (PDA), an MP3 player, or the like.

[0117] Although a display device, a controller, and a method of driving the display device according to embodiments have been described with reference to the drawings, embodiments are examples, and may be modified and changed by a person having ordinary knowledge in the relevant technical field without departing from the technical spirit described in the following claims.

Claims

1. A display device, comprising: a display panel including a plurality of pixels; a power management circuit which provides a power voltage to the plurality of pixels through a power line; a current sensing circuit which senses a panel current flowing through the power line and performs an analog-to-digital conversion for converting the panel current into a panel current code; and a controller which sets at least one of a conversion range, a conversion time, or a conversion sampling count of the analog-to-digital conversion based on a dimming value for controlling a luminance of an image displayed by the display panel.

2. The display device of claim 1, wherein the controller sets the conversion range such that the conversion range increases as the dimming value increases and the conversion range decreases as the dimming value decreases, sets the conversion time such that the conversion time decreases as the dimming value increases and the conversion time increases as the dimming value decreases, and sets the conversion sampling count such that the conversion sampling count decreases as the dimming value increases and the conversion sampling count increases as the dimming value decreases.

3. The display device of claim 1, wherein the dimming value includes a dimming gain value for decreasing the luminance of the image in a total load section and a dimming peak value for decreasing the luminance of the image in a local load section.

4. The display device of claim 3, wherein the controller sets the conversion range such that the

conversion range increases as the dimming gain value increases and the conversion range decreases as the dimming gain value decreases.

5. The display device of claim 3, wherein the controller sets the conversion time such that the conversion time decreases as the dimming gain value increases and the conversion time increases as the dimming gain value decreases.

6. The display device of claim 3, wherein the controller sets the conversion sampling count such that the conversion sampling count decreases as the dimming gain value increases and the conversion sampling count increases as the dimming gain value decreases.

7. The display device of claim 3, wherein the controller sets the conversion time such that the conversion time decreases as the dimming peak value increases and the conversion time increases as the dimming peak value decreases.

8. The display device of claim 3, wherein the controller sets the conversion sampling count such that the conversion sampling count decreases as the dimming peak value increases and the conversion sampling count increases as the dimming peak value decreases.

9. The display device of claim 3, wherein the controller includes: a net power control block which determines a load of input image data and a scale factor based on the input image data, a net power control lookup table, and the dimming value; and a conversion setting block of the controller, which sets the at least one of the conversion range, the conversion time, or the conversion sampling count based on the dimming value, and provides the at least one of the conversion range, the conversion time, or the conversion sampling count to the current sensing circuit.

10. The display device of claim 9, wherein the conversion setting block includes: a conversion range setting block which sets the conversion range based on the dimming gain value; a conversion time setting block which sets the conversion time based on the dimming gain value and the dimming peak value; and a conversion sampling count setting block which sets the conversion sampling count based on the dimming gain value and the dimming peak value.

11. The display device of claim 9, wherein the controller further includes: a global current management block which receives the panel current code from the current sensing circuit, determines a target current code corresponding to the load of the input image data, and controls the power management circuit to supply the panel current according to a comparison of the panel current code with the target current code.

12. The display device of claim 9, wherein the controller further includes: an over-current protection block which receives the panel current code from the current sensing circuit, compares the panel current code with a reference over-current code, and controls the power management circuit to stop supplying the power voltage when the panel current code is greater than the reference over-current code.

13. The display device of claim 1, wherein the current sensing circuit includes: a conversion block which converts the panel current into the panel current code; and a setting register which stores the conversion range, the conversion time, and the conversion sampling count.

14. The display device of claim 13, wherein the controller is implemented as a controller integrated circuit, and wherein the current sensing circuit is implemented as a current sensing integrated circuit different from the controller integrated circuit.

15. The display device of claim 14, wherein the controller sets the at least one of the conversion range, the conversion time, or the conversion sampling count by changing a value of the setting register via a bus connecting between the controller integrated circuit and the current sensing integrated circuit.

16. A controller of a display device, comprising: a net power control block which determines a scale factor based on input image data, a net power control lookup table, and a dimming value for controlling a luminance of an image displayed by the display device; and a conversion setting block which sets at least one of a conversion range, a conversion time, or a conversion sampling count of an analog-to-digital conversion for converting a panel current into a panel current code

based on the dimming value, and provides the at least one of the conversion range, the conversion time, or the conversion sampling count to a current sensing circuit which senses the panel current.

17. The controller of claim 16, wherein the conversion setting block sets the conversion range such that the conversion range increases as the dimming value increases and the conversion range decreases as the dimming value decreases, sets the conversion time such that the conversion time decreases as the dimming value increases and the conversion time increases as the dimming value decreases, and sets the conversion sampling count such that the conversion sampling count decreases as the dimming value increases and the conversion sampling count increases as the dimming value decreases.

18. The controller of claim 16, wherein the controller is implemented as a controller integrated circuit, wherein the current sensing circuit is implemented as a current sensing integrated circuit different from the controller integrated circuit, and wherein the conversion setting block provides the at least one of the conversion range, the conversion time, or the conversion sampling count to the current sensing circuit via a bus connecting between the controller integrated circuit and the current sensing integrated circuit.

19. A method of driving a display device, the method comprising: setting at least one of a conversion range, a conversion time, or a conversion sampling count of an analog-to-digital conversion for converting a panel current into a panel current code based on a dimming value for controlling a luminance of an image displayed by the display device; sensing a panel current; and performing the analog-to-digital conversion for converting the panel current into the panel current code.

20. The method of claim 19, wherein the setting comprises at least one of setting the conversion range such that the conversion range increases as the dimming value increases and the conversion range decreases as the dimming value decreases, setting the conversion time such that the conversion time decreases as the dimming value increases and the conversion time increases as the dimming value decreases, or setting the conversion sampling count such that the conversion sampling count decreases as the dimming value increases and the conversion sampling count increases as the dimming value decreases.
